

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0810

Easton 2 345 kV
June 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0810. The Interconnection Customer proposes a 437.9 MW wind facility interconnecting at the Easton 2 345 kV substation in the Maryland area. The request was received on June 16, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 243 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,049,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NAVY MOLE – EASTON 2 138kV	CADY – EASTON 2 345kV	264.5	412	54.9
EASTON 2 345/138 kV Transformer	CADY – EASTON 2 345kV	261.7	416	54.9
PARK AVE – HILLSBORO SUBSTATION 138kV	CADY – EASTON 2 345kV line	222.8	296	25.9
HOLSTON DAM – NAVY MOLE 138kV	CADY SUBSTATION – EASTON 2 345kV	211.8	390	31.5
GRASONVILLE – MACON 138kV Line	CADY – EASTON 2 345kV	191.3	197	14.7
CANNELTON HYDROELECTRIC PLANT SUBSTATION – EQUUS FREEPORT POWER 138kV	CADY – EASTON 2 345kV	162.9	259	12.6
KNIGHT STREET – EQUUS FREEPORT POWER SUBSTATION 138kV	CADY – EASTON 2 345kV Line	158.0	279	12.6
CANNELTON HYDROELECTRIC PLANT – HENRY ST WPS 138kV	CADY – EASTON 2 345kV L	157.4	248	12.6
KNIGHT STREET – LONG LAKE 138kV	CADY SUBSTATION – EASTON 2 345kV	155.3	303	12.6
LINDE – RIPPLE 138kV	CADY – EASTO 2 345kV	146.8	337	12.6
MCMINNVILLE SUBSTATION – LONG LAKE 138kV	EASTON 2 – CADY 345kV	145.5	329	12.6
LINDE SUBSTATION – MCMINNVILLE 138kV	EASTON 2 – CADY 345kV	143.6	339	12.6
HENNING – KIWI 138kV L	EASTON 2 – CADY 345kV	142.5	262	14.1
QUINN – HENRY ST WPS 138kV	EASTON 2 – CADY SUBSTATION 345kV	140.7	262	12.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
YANKE – JARRATT 138kV	CADY – EASTON 2 345kV	118.6	285	9.1
YANKE – PARK AVE 138kV L	CADY – EASTON 2 345kV	112.4	295	9.1
GRASONVILLE – CARANCAHUA 138kV Line	CADY SUBSTATION – EASTON 2 345kV	100.6	196	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NAVY MOLE – EASTON 2 138kV	Rebuild line with higher-capacity conductor	\$467,000	19% (\$89,000)
EASTON 2 345/138 kV Transformer	Replace transformer with higher-rated unit	\$17,151,000	19% (\$3,258,000)
PARK AVE – HILLSBORO 138kV Line	Rebuild line with higher-capacity conductor	\$8,853,000	15% (\$1,345,000)
HOLSTON DAM – NAVY MOLE 138kV	Rebuild line with higher-capacity conductor	\$2,830,000	19% (\$528,000)
GRASONVILLE – MACON 138kV Line	Rebuild line with higher-capacity conductor	\$2,127,000	17% (\$370,000)
CANNELTON HYDROELECTRIC PLANT – EQUUS FREEPORT POWER 138kV line	Rebuild line with higher-capacity conductor	\$1,529,000	12% (\$186,000)
KNIGHT STREET – EQUUS FREEPORT POWER SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$2,839,000	12% (\$345,000)
CANNELTON HYDROELECTRIC PLANT – HENRY ST WPS 138kV	Rebuild line with higher-capacity conductor	\$4,182,000	12% (\$508,000)
KNIGHT STREET – LONG LAKE 138kV	Rebuild line with higher-capacity conductor	\$8,262,000	12% (\$1,004,000)
LINDE – RIPPLE 138kV	Rebuild line with higher-capacity conductor	\$7,932,000	12% (\$964,000)
MCMINNVILLE SUBSTATION – LONG LAKE 138kV	Rebuild line with higher-capacity conductor	\$13,870,000	12% (\$1,686,000)
LINDE SUBSTATION – MCMINNVILLE 138kV	Rebuild line with higher-capacity conductor	\$5,738,000	12% (\$697,000)
HENNING – KIWI 138kV L	Rebuild line with higher-capacity conductor	\$3,376,000	14% (\$489,000)
HENRY ST WPS – QUINN 138kV	Rebuild line with higher-capacity conductor	\$2,586,000	12% (\$314,000)
YANKE – JARRATT 138kV	Rebuild line with higher-capacity conductor	\$11,273,000	11% (\$1,194,000)
YANKE – PARK AVE 138kV Line	Rebuild line with higher-capacity conductor	\$4,901,000	11% (\$519,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRASONVILLE – CARANCAHUA 138kV Line	Reconductor line	\$3,055,000	18% (\$553,000)
	Total allocated to project		\$14,049,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.